

Unit - II

Plane Trusses

1.) Define Truss, members and joints.

* Truss is an assembly of structural members assembled by joints.

* Members is the function of carrying the loads.

* Joints is nothing but connection of two or more members by means of rivets, welds, bolts etc.

2.) Classify the Trusses.

Trusses are classified based on

1. Dimensional Configuration

a.) Plane Truss

b.) Space Truss

2. Determinacy

a.) statically determinate truss

b.) statically indeterminate truss.

3. Supports.

3.) Differentiate between Plane truss and Space truss.

Plane Truss: A truss of which the members lies in a single plane is called as plane truss. Eg. Roof Truss, Bridge Truss.

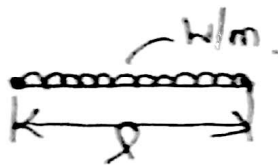
Space Truss: A Truss of which the members lies in different planes is called as space truss. Eg: Tripod, Transmission towers.

4.) what are the different types of supports?

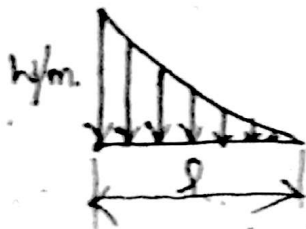
1. Hinged support (Pin-joint support)
2. Roller support
3. Fixed support.

5.) what are the different types of loads?

1. Point load. 
2. Uniformly distributed load (UDL)



3. Uniformly varying load (UVL)



6.) what are the different types of beams?

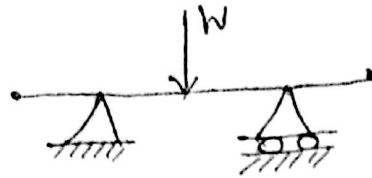
1. Simply supported beam



2. Cantilever beam.



3. Overhanging beam



7.) What is a statically determinate Truss ?

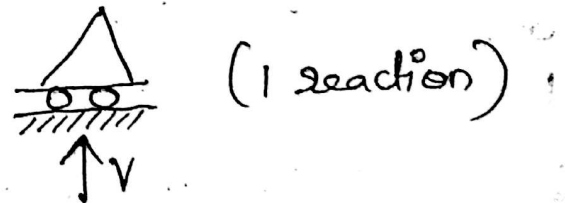
A truss which could be analysed completely using the three static equilibrium equations alone ($\sum H=0$, $\sum V=0$ and $\sum M=0$) is called as statically determinate Truss.

8.) What is statically indeterminate Truss ?

A truss which could not be analysed by the static equilibrium equations alone ($\sum H=0$, $\sum V=0$ & $\sum M=0$) for the complete analysis needs additional equations is called as statically indeterminate truss.

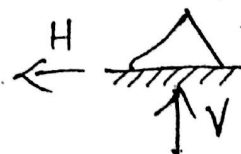
9.) Draw the reaction components of Roller, hinged and Fixed support.

1. Roller Support



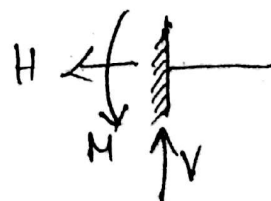
(1 reaction)

2. Hinged support



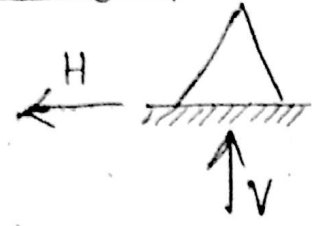
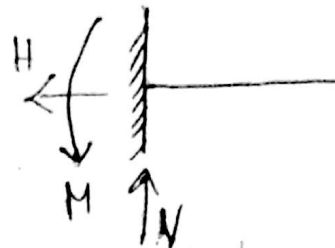
(2 reactions)

3. Fixed support

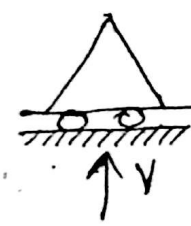
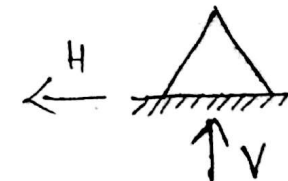


(3 reactions)

10.) What is the difference between a hinged support and a fixed support?

No.	Hinged Support	Fixed Support
1.)	 Hinged Support	 Fixed Support
2.)	This type of support can withstand any type of force both horizontal & vertical.	Fixed Support has three reaction components, horizontal, vertical & rotational reaction. Fixed support is considered as the strongest support.

11.) Compare roller support and hinged support.

S.No	Roller Support	Hinged Support
1.		
2.	Roller Support has the known line of action of reaction	Hinged Support has an unknown line of action of reaction.
3.	Roller Support has only one reaction.	Hinged Support has two reactions.

12.) What is the condition of perfect and imperfect truss?

The condition of Perfect truss is

$$m = 2j - 3$$

where,

m = no. of members in truss

j = no. of joints in truss.

The condition of Imperfect truss is

$$m \neq 2j - 3$$

13.) Define friction & its types.

When two surfaces are in contact with each other and one surface tends to move with respect to the other, a tangential force will be developed at the contact surface in the opposite direction of motion. This tangential force is called frictional force or simply friction.

In general, two types of friction can occur between surfaces, namely

- (i) Dry friction (or Coulomb friction)
- (ii) fluid friction.

14.) Define limiting friction.

Limiting friction is defined as the product of coefficient of friction & normal reaction.

$$\text{Limiting friction} = \text{Coefficient of friction} \times \text{normal reaction.}$$

15.) Define angle of repose.

"The angle of the inclined plane, at which the body tends to slide down is known as angle of repose."

$$\alpha_m = \phi$$

16.) Define Co-efficient of friction.

The ratio of limiting friction to the normal reaction is known as Co-efficient of friction. It is denoted by the symbol μ .

$$\mu = \frac{\text{Limiting friction}}{\text{Normal reaction}} = \frac{F_m}{N_R}$$

17.) Define angle of friction

The angle between the resultant R and the normal reaction N_R is called angle of friction. It is denoted by the symbol ϕ .

$$\tan \phi = \frac{F_m}{N_R}$$

18.) Define Dry friction?

Dry friction refers to the friction which develops between two dry surfaces, slide or tends to slide relative to another.

19.) What do you mean by impending motion?

The state of motion of a body which is just about to move or slide is called impending motion. When the maximum frictional force is attained and if the applied force exceeds the limiting friction, the body starts sliding or rolling. This state is called impending motion.

20.) Differentiate between co-efficient of static & dynamic friction.

Co-efficient of friction in static state is called co-efficient of static friction (μ_s).

Co-efficient of friction in dynamic state is called co-efficient of dynamic friction (μ_k).

21.) Define cone of friction.

The inverted cone with semi-conical angle equal to the angle of friction is called cone of friction.

22.) Write an expression for the torque required to work the screw-jack lifting a load W .

$$P = \frac{W \cdot r}{L} \tan(\phi + \alpha)$$

$$Q = \frac{W \cdot r}{L} \tan(\phi - \alpha)$$

23.) Define Rolling resistance & co-efficient of Rolling resistance.

When one body is made to roll freely over another body, a resistance is developed in the opposite direction known as Rolling resistance.

Horizontal distance of point of resistance measured from centre of wheel known as co-efficient of Rolling resistance.

Q4.) State Coulomb's law of dry friction?

(i) Laws of static friction:

(i) The frictional force always acts in the opposite direction to that the body tends to move.

(ii) The frictional force does not depend on the shape and area of contact of the bodies.

(ii) Laws of dynamic friction:

(i) The frictional force always acts in the opposite direction to that the body moves.

(ii) The co-efficient of kinetic friction is less than the co-efficient of static friction.

Unit III

Properties of Surfaces

1. Distinguish between Centroid and Centre of gravity?

The centre of gravity of a body is defined as the Point through which the entire weight of the body acts; when this is referred to weightless laminae or plane areas is called the Centroid of the area.

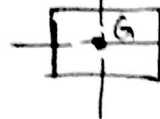
2. Distinguish between Reference axes and Centroidal axes.

⇒ Location of Centroid is always measured with reference to some reference axes. Generally two axes are used for the Centroid of a plane figures (i.e) OX and OY .



Reference axes

⇒ The axes which are passing through the Centroid of the figures are called the Centroidal axes. The horizontal and vertical Centroidal axes passing through G are xx and yy axes.



Centroidal axes

3.) What are the methods to determine Centre of gravity?

(1) By Geometrical Consideration

(2) By Graphical method

(3) By M. of moments

4.) By Integration

4.) Define moment of inertia (or) second moment of inertia?

The product of area and square of the distance from reference axis is called Second moment of area about the reference axis. Second moment of area is also known as Moment of inertia. It is represented by 'I'.
Second Moment of Area about $xx = I_{xx} = A \bar{y}^2$.
Second Moment of area about $yy = I_{yy} = A \bar{x}^2$.

5.) Define first moment of inertia?

Moment of an area about a point is the product of area and its centroidal distance from the point.

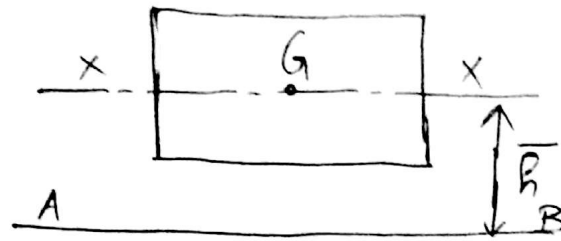
The moment of area divided by area will give the centroid of the area. This is called as first moment of Area or first moment of inertia.
$$\bar{x} = \frac{\sum a \bar{x}}{\sum a} = \frac{\text{Sum of moment of Areas about } OY}{\text{Total Area.}}$$

6.) Define axis of Symmetry?

If a composite plane figures has an axis of symmetry (i.e. an axis about which similar configuration is seen on either side), Centroid lies on it. This concept reduce the work of locating the centroid.

7.) State Parallel axis theorem.

It states that "the moment of inertia of a lamina about any axis in the plane of lamina is equal to the sum of the moment of inertia about a parallel centroidal axis in the plane of lamina and the Product of the area of the lamina and Square of the distance between the two axes".



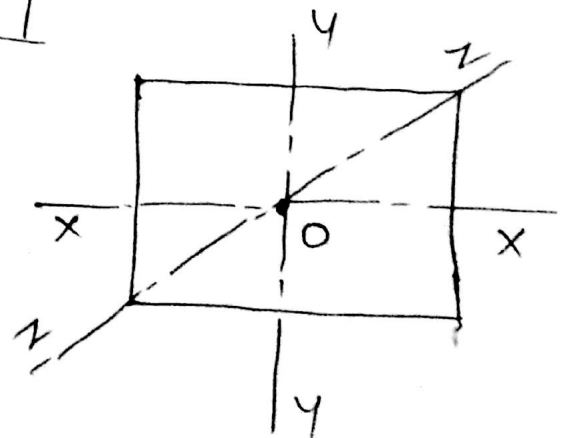
$$I_{AB} = I_{xx} + Ah^2$$

8.) State Perpendicular axis theorem.

It states that "If I_{ox} and I_{oy} be the moment of inertia of a lamina about two mutually Perpendicular axes ox and oy in the plane of the lamina and I_{oz} be the moment of inertia of the lamina about an axis normal to the lamina and passing through the Point of Intersection of the axes ox and oy , then

$$I_{oz} = I_{ox} + I_{oy}$$

$$I_{zz} = I_{xx} + I_{yy}$$



9.) Define Polar moment of Inertia.

The area moment of Inertia (i.e. M.I of plane figures) for an area relative to an axis perpendicular to the plane of the area is called the Polar moment of Inertia. It is denoted by I_p .

$$I_p = I_{xx} + I_{yy}$$

10.) Define Radius of gyration.

Radius of gyration (also called as radius of rotation) about an axis is defined as the distance from that axis at which all the elemental parts of the lamina would have to be placed (i.e. the entire area A of the figure is assumed to be concentrated), such that the moment of Inertia about the axis is same.

$$r = \sqrt{\frac{I}{A}}$$

where,

r = radius of gyration

I = moment of Inertia

A = area of the section

$$r_{xx} = \sqrt{\frac{I_{xx}}{A}}, \quad r_{yy} = \sqrt{\frac{I_{yy}}{A}}$$

11) Define Product of Inertia. [Product moment of Inertia].

Product of Inertia is defined as Product of the distances of the area from the axis under consideration and the element of the area.

$$\text{Hence, } I_{xy} = \int_A xy dA.$$

12) Define mass moment of Inertia.

The product of the mass and the square of the distance of the centre of gravity of the mass from an axis is known as the mass moment of inertia about that axis.

$$\therefore I_m = M r^2.$$

13) State Pappus and Guldinus theorem.

Theorem 1: It states that "the area of a surface of revolution is the product of the length of the generating curve and the distance travelled by the centroid of the curve, while the surface is generated".

Theorem 2: It states that "the volume of a body of revolution is obtained from the product of the generating

area and the distance travelled by the centroid of the area, while the body is being generated.

14.) Write the expressions to find the Centroid of a Simple Plane Figures

$$\bar{x} = \frac{a_1x_1 + a_2x_2 + \dots + a_nx_n}{a_1 + a_2 + \dots + a_n}$$

$$\bar{y} = \frac{a_1y_1 + a_2y_2 + \dots + a_ny_n}{a_1 + a_2 + \dots + a_n}$$

where,

$a_1, a_2, \dots, a_n$ - area of simple plane figures 1, 2, ..., of the composite plane figure.

$x_1, x_2, \dots, x_n$ - horizontal distance of the centroid of simple plane figures 1, 2, ...

$y_1, y_2, \dots, y_n$ - vertical distance of the centroid of simple plane figures 1, 2, ...

15.) Write the expression to determine Centre of gravity of a Composite Solid figures.

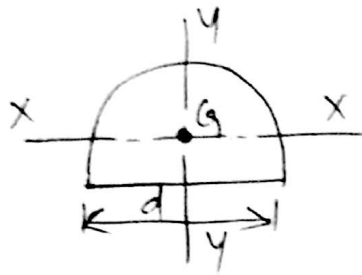
$$\bar{x} = \frac{W_1x_1 + W_2x_2 + \dots + W_nx_n}{W_1 + W_2 + \dots + W_n}$$

$$\bar{y} = \frac{w_1 y_1 + w_2 y_2 + \dots + w_n y_n}{w_1 + w_2 + \dots + w_n}$$

where,

w_1, w_2 are the weight of the components 1 & 2.

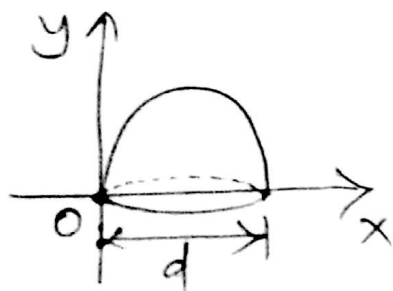
16.) What is the moment of inertia of a semicircle about its base?



$$I_{xx} = 0.0068d^4$$

$$I_{yy} = \frac{\pi d^4}{128}$$

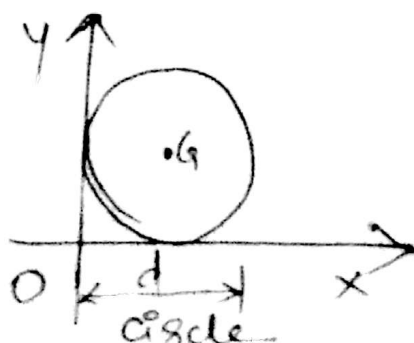
17.) Write the expression for centroid of a hemisphere and circular lamina?



Hemisphere.

$$\bar{x} = \frac{d}{2}$$

$$\bar{y} = \frac{3r}{8}$$



circle

$$\bar{x} = \frac{d}{2}$$

$$\bar{y} = \frac{d}{2}$$

Kinematics and Kinetics of Particles

1.) Differentiate between kinematics and kinetics.

Kinematics is the study of motion of bodies without considering the forces causing motion.

Kinetics deals with the forces causing motion.

2.) List out different types of motion.

- Rectilinear motion
- Curvilinear motion.

3.) Define the average velocity.

It is the ratio of displacement and time interval.

$$(i.e) \text{ average velocity} = \frac{\text{Displacement}}{\text{time taken.}}$$

4.) Differentiate Rectilinear motion and Curvilinear motion.

Sl.no.	Rectilinear motion	Curvilinear motion.
1.	The motion of a particle along a straight line is known as rectilinear motion (straight line motion). Example: A car moving on a straight road.	The motion of a particle along a curved path is known as Curvilinear motion.

5.) Define relative motion.

When the motion of a body is evaluated with respect to another moving body, then it is called as relative motion.

6.) Differentiate between speed and velocity.

The rate of change of distance with respect to time is called speed whereas the rate of change of displacement with respect to time is called velocity. Speed has the magnitude only, whereas velocity has both magnitude and direction. Speed is a scalar quantity. Velocity is a vector quantity.

7.) Differentiate between Acceleration and retardation.

Rate of change of velocity with respect to time is called acceleration.

$$a = \frac{dv}{dt}.$$

The positive acceleration is simply referred as acceleration and the negative acceleration is called as retardation.

8.) State Newton's law.

Newton's first law: Everybody continues in its state of rest or of uniform motion in a straight line unless it is compelled to change that state by forces acting on it.

Newton's second law: "The rate of change of momentum of a body is directly proportional to the force acting on it and takes place in the direction of force."

$$\boxed{F = ma}$$

Newton's third law: To every action, there is an equal and opposite reaction.

9.) What is Instantaneous Velocity?

The instantaneous velocity of a particle at any instant of time is the limit of average velocity as the increment of time approaches zero.

$$\boxed{v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}}$$

10.) Write the equations of motion of a particle under gravity.

Vertical downward motion

$$v = u + gt$$

$$v^2 = u^2 + 2gh$$

$$h = ut + \frac{1}{2}gt^2$$

Vertical upward motion

$$v = u - gt$$

$$v^2 = u^2 - 2gh$$

$$h = ut - \frac{1}{2}gt^2$$

11.) Define normal and tangential acceleration?



Components of acceleration

The acceleration at any point along the tangent. These components are called tangential (a_t) acceleration. It is due to change in magnitude of velocity.

The acceleration at any point along normal is called normal acceleration. It is due to change in direction of velocity (a_n).

12.) Define Projectile.

A Particle projected in space at an angle to the horizontal plane is called as Projectile.

13.) Define angle of Projection.

The angle to the horizontal, at which the Projectile is projected is called angle of Projection.

14.) Define velocity of projection.

The velocity with which the Particle is projected is called as velocity of projection (u) m/s.

15.) Define Trajectory.

The path traced by the Projectile is called Trajectory.

16.) Define time of flight.

The time interval during which the Projectile is in motion is called time of flight.

17.) Define Horizontal range (Range).

It is the distance along the plane between the point of projection and the point at which the Projectile hits the plane at the end of its journey.

18.) State D'Alembert's Principle.

It states that "the system of forces acting on a body in motion is in dynamic equilibrium, with the inertia force of the body".

$$\therefore -F - ma = 0$$

19.) What is work-energy Principle.

The work energy Principle may be stated as the work done by a system of forces acting on a body during a displacement is equal to the change in kinetic energy of the body during the same displacement.

$$\boxed{F \cdot S = \frac{W}{2g} (v^2 - u^2)}$$

→ work energy equation

20.) Write impulse momentum equation.

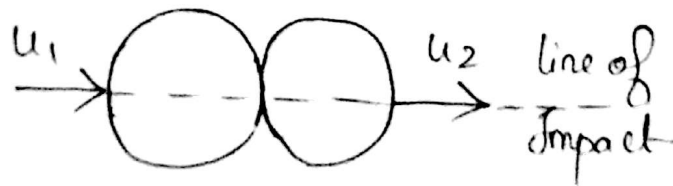
The impulse of the force acting on a particle is equal to the change in the linear momentum of the particle.

Impulse = final momentum - initial momentum.

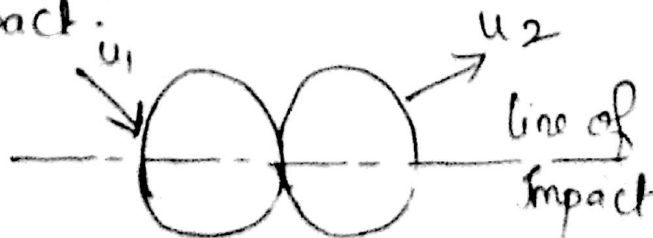
$$\text{Impulse} = m(v - u).$$

21.) Differentiate direct and oblique impact.

In direct impact, the velocities of the two colliding bodies before collision are collinear with the line of impact.



In oblique impact, the velocities of the two colliding bodies before collision are not collinear with the line of impact.



22.) Differentiate direct Central Impact and oblique Central Impact.

In direct Central Impact, the velocities of the colliding bodies are directed along the line of impact and the line of impact passes through the mass centres of the colliding bodies.

In oblique Central Impact, the velocities of the colliding bodies are not directed along the line of impact and the line of impact passes through the mass centres of the colliding bodies.

23.) Define Co-efficient of restitution.

The ratio of the magnitude of restitution impulse to the magnitude of deformation impulse is known as Co-efficient of restitution denoted by the letter 'e'.

$$\text{Co-efficient of restitution} = \frac{\text{Impulse during restitution}}{\text{Impulse during deformation}}$$

$$e = \frac{\int R dt}{\int P dt}$$

24.) Define period of restitution.

The time elapsed from the instant of initial contact to the maximum deformation is known as Period of deformation and the time elapsed from the instant of maximum deformation to the instant of just separation of particles is known as Period of restitution.

25.) State law of conservation of momentum.

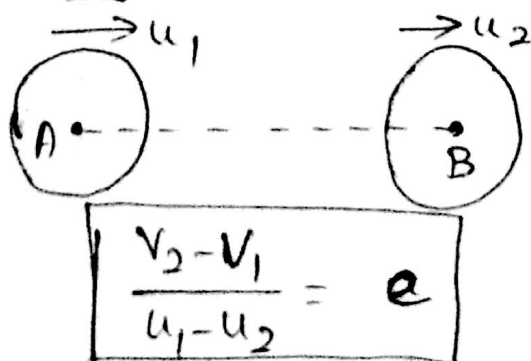
When two elastic bodies collide each other, the momentum gained by one body must be equal to the momentum lost by the other body." i.e. net change in the momentum of the system is zero.

Total momentum before impact = Total momentum after impact.

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

26.) State Newton's law of collision of elastic bodies.

It states that "for two colliding bodies, their relative velocity of separation bears a constant ratio to their relative velocity of approach".



where,
 e = co-efficient of restitution (or) Co-efficient of elasticity.

Kinematics & Kinetics of Rigidbodies

1. Define Plane motion.

If each particle of the body remains at a constant distance from a fixed reference plane during the motion, it is said to be in plane motion.

2. What are the types of Plane rigid body motion?

1. Translation

a.) Rectilinear translation

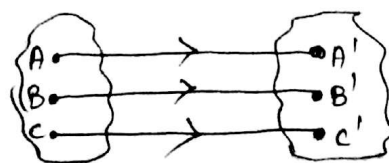
b.) Curvilinear translation

2. Rotation

3. General plane motion.

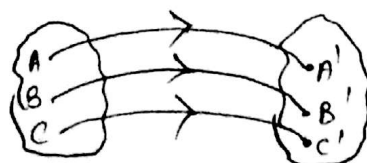
3. Differentiate between rectilinear & curvilinear translation.

Rectilinear translation: The motion of a rigidbody along a straightline is known as rectilinear translation.



Rectilinear translation.

Curvilinear translation: The motion of a rigidbody along a curved path is known as curvilinear translation.



Curvilinear translation

4. Define general plane motion.

A rigid body is said to be in general plane motion, if it undergoes a combination of translation and rotation.

Example,

- (i) Rolling wheel
- (ii) Sliding rod.

5. Define angular displacement and angular velocity.

angular displacement: It is defined as the total angle, through which a body has rotated. It is denoted by θ .

$$\theta = f(t)$$

angular velocity: It is the rate of change of angular displacement of the body. It is denoted by ω .

$$\omega = \frac{d\theta}{dt}$$

6. Define angular acceleration?

Angular acceleration: It is the rate of change of angular velocity. It is denoted by α .

$$\alpha = \frac{d\omega}{dt}$$

7.) Write the equation of motion of rigid body, rotating with uniform angular acceleration.

$$\omega = \omega_0 + \alpha t$$

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

8.) State Chasle's theorem.

Chasle's theorem states that "the most general motion of a rigid body is equivalent to a translation of the body with some point of the body plus a rotation of the body about an axis passing through that point, and the order of such translation and rotation is immaterial".

9.) Define Instantaneous Centre of rotation.

The combined motion of translation and rotation may be assumed to be a motion of pure rotation about a centre point is known as Instantaneous Centre of rotation. It is denoted by I_c .

10.) Write the equation for D'Alembert's principle for rigid body.

$$\sum F_x - ma_x = 0$$

$$\sum F_y - ma_y = 0$$

$$\sum M_G - I_G \alpha = 0$$

11.) write the equation of work-energy method for rigid body.?

$$\Sigma M_G \theta = \frac{1}{2} I_G (\omega^2 - \omega_0^2)$$

12.) write the equation of impulse-momentum for rigid body?

$$\Sigma M_G t = I_G (\omega - \omega_0)$$

13.) Define constrained and unconstrained motion.

Constrained motion: If the motion of the body is confined or defined then the motion is called constrained motion. There are two types.

(i) Point constraint: The rotation of the rigid body about a fixed point or axis.

(ii) Line constraint: Motion of a body along a straight line.

unconstrained motion:

If there is no confinement to the motion then the motion is called unconstrained motion.

eg: motion of a bus, car, rockets.

14.) Define Centre of Percussion?

Centre of Percussion is a Point through which the resultant of the effective forces act for a rigid body. It follows that the sum of the moments of all forces about the Centre of Percussion is always zero.

$$\boxed{\sum M_G = 0}$$

15.) Define Coriolis's acceleration?

For the acceleration of a rigid body, having both transverse and rotational motion, the acceleration are

$$a_r = \frac{d^2 r}{dt^2} - r \left(\frac{d\theta}{dt} \right)^2$$

$$a_n = r \frac{d^2 \theta}{dt^2} + 2 \left(\frac{dr}{dt} \times \frac{d\theta}{dt} \right)$$

The term $2 \left(\frac{dr}{dt} \times \frac{d\theta}{dt} \right) = 2v\omega$ is called Coriolis's acceleration.

16.) A flywheel of radius 1.5m is rotating with a speed of 1000 rpm. Calculate the angular velocity & linear velocity.

Given:

$$r = 1.5m \quad N = 1000 \text{ rpm}$$

$$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 1000}{60} = 104.71 \text{ rad/sec.}$$

To find:

Soln:

$$\omega = ? \quad v = ?$$

$$\boxed{\omega = 104.71 \text{ rad/sec}}$$

$$\omega = \frac{v}{r}$$

$$104.71 = \frac{v}{1.5}$$

$$\Rightarrow \boxed{v = 157 \text{ m/s}}$$

Result:

$$\omega = 104.71 \text{ rad/sec}, \quad v = 157 \text{ m/s}$$

17.)

A body is rotating with an angular velocity of 5. After 4 seconds, the angular velocity of the body becomes 13 rad/sec. Determine the angular acceleration of the body.

Given:

$$\omega_0 = 5 \text{ rad/sec}$$

$$\omega = 13 \text{ rad/sec.}$$

$$t = 4 \text{ sec}$$

To find

$$\alpha = ?$$

Soln:

W.K.T

$$\omega = \omega_0 + \alpha t$$

$$13 = 5 + \alpha(4)$$

$$\boxed{\alpha = 2 \text{ rad/sec}^2}$$

Result:

18.) A wheel rotating about a fixed axis at 20 rpm is uniformly accelerated for 70 sec, during which time it takes 50 revolutions. Find the final angular velocity of the wheel.

Soln:

$$N_0 = 20 \text{ rpm}$$

$$\omega_0 = \frac{2\pi N}{60} = \frac{2\pi \times 20}{60}$$

$$\omega_0 = 2.094 \text{ rad/sec}$$

$$t = 70 \text{ sec}$$

$$\theta = 50 \text{ rev} = 50 \times 2\pi = 314.15 \text{ rad}$$

W.K.T

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$314.15 = 2.094(70) + \frac{1}{2} \alpha (70)^2$$

$$\boxed{\alpha = 0.068 \text{ rad/sec}^2}$$

$$\omega = \omega_0 + \alpha t$$

$$\omega = 2.094 + (0.068)(70)$$

$$\Rightarrow \boxed{\omega = 6.854 \text{ rad/sec}}$$

Result:

$$\alpha = 0.068 \text{ rad/sec}^2$$

$$\omega = 6.854 \text{ rad/sec}$$

17.) The earth makes a complete revolution about its own axis in 24 hours. If the velocity at any point on its equator is 1675 km per hour, find the radius of the earth in kilometres.

Given

$$\theta = 1 \text{ rev (a complete rev.)}$$

$$\theta = 2\pi \text{ rad. } \{ \because 1 \text{ rev} = 2\pi \}$$

$$t = 24 \text{ hours} = 24 \times 60 \times 60 = 86400 \text{ sec.}$$

$$V = 1675 \text{ km/hr} = \frac{1675 \times 1000}{3600} = 465.277 \text{ m/s}$$

To find

$$r = ?$$

Soln

$$\omega = \frac{\theta}{t} = \frac{2\pi}{86400} = 7.272 \times 10^{-5} \text{ rad/sec}$$

$$\omega = \frac{V}{r} = \frac{465.277}{r}$$

$$7.272 \times 10^{-5} = \frac{465.277}{r}$$

$$r = 6398018.017 \text{ m}$$

$$r = 6398.018 \text{ km}$$

Result:

$$r = 6398.018 \text{ km}$$

20.) State linear impulse and momentum.

Linear Momentum: Product of mass and velocity is called as linear momentum.

$$\text{momentum} = mv.$$

Linear impulse:

When a large force acts for a short period of time, that force is called as impulsive force.

$$\text{Linear impulse} = \text{force} \times \text{Time}.$$

both impulse & momentum are measured in N-sec.

21.) State angular impulse and momentum.

Angular momentum: angular momentum is the product of mass moment of inertia and angular velocity.

$$\text{angular momentum} = I \times \omega$$

Angular impulse

Product of moments and time.

$$\text{Angular impulse} = M \times t.$$